- 4) Apply concepts of additive manufacturing to determine part build orientation.
- 5) Apply concepts of additive manufacturing to design support structures.
- 6) Integrate concepts of part design, orientation, and support structures to apply a holistic design strategy to additive manufacturing.
- 7) Differentiate additive manufacturing technologies as functions of their design considerations.

**AMFG598. SPECIAL TOPICS IN ADVANCED MANUFACTURING. 1-6 Semester Hr.**
(I, II) Pilot course or special topics course. Topics chosen from special interests of instructor(s) and student(s). Usually the course is offered only once. Prerequisite: none. Variable credit; 1 to 6 credit hours. Repeatable for credit under different titles.

**AMFG598. SPECIAL TOPICS IN ADVANCED MANUFACTURING. 1-6 Semester Hr.**

**AMFG599. INDEPENDENT STUDY. 1-6 Semester Hr.**

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**AMFG599. INDEPENDENT STUDY. 1-6 Semester Hr.**

## Director and Professor of Practice

Craig A. Brice

# Carbon Capture, Utilization and Storage

## Degrees Offered

- Master of Science (Non-Thesis) in Carbon Capture, Utilization and Storage
- Carbon Capture, Utilization and Storage Certificate

## Program Description

As global interest in carbon capture, utilization, and storage (CCUS) continues to grow, so does the demand for skilled professionals equipped to tackle its unique challenges. The Mines CCUS graduate certificate program addresses this critical need. By integrating world-renowned expertise in earth sciences, engineering, economics and business, this interdisciplinary program enables students to explore the unique challenges related to CCUS, climate change, and the energy transition while gaining the specialized skills needed to advance their careers.

The Masters in Carbon Capture, Utilization and Storage (course-based) program is designed to give students a foundation in the technical, business, and community engagement aspects of CCUS to help them further their careers in the CCUS space. This non-thesis, online only program is structured to be of use for both those seeking to expand their career or those just getting started.

The CCUS graduate certificate program is designed for professionals looking to enhance their expertise in carbon capture, utilization, and storage. The courses are fully asynchronous, eight weeks long, and offered online, providing flexibility for working professionals. Students need to complete three courses - two required and one elective - to earn the graduate certificate in CCUS. These courses can also be taken to count towards a graduate degree.

Participants in this program can expect to:

- Assess data on climate change, and the effects of greenhouse gases on climate.
- Develop a solid foundation and proficiency in methods employed for the three aspects of carbon capture, utilization, and storage.
- Learn workflow and practices in the industry. Assess efficiency of CCUS practices.
- Develop skills to better communicate with colleagues in other disciplines in the organization.
- Learn and understand fundamental concepts from well-known, experienced faculty in the field.

## Primary Contact

Program Director: Yilin Fan

https://ccus.mines.edu

## Masters in Carbon Capture, Utilization and Storage

The program offers two degree options in Carbon Capture, Utilization, and Storage (CCUS): a Graduate Certificate and a Master of Science (Non-Thesis).

The Master of Science (Non-Thesis) in CCUS is a fully online degree that provides students with an interdisciplinary foundation in the science, technology, and policy dimensions of carbon capture, utilization, and storage. The program requires the completion of 30 graduate credits, consisting of three core courses and seven electives. The core courses introduce fundamental perspectives on climate change and sustainability, the political economy of the energy transition, and include a capstone project. Elective courses allow students to tailor the program to their interests, with options spanning geological and non-geological CCUS, biological carbon capture and conversion, CO# reaction kinetics, Class VI well design and permitting, and related topics. This flexible curriculum equips graduates to address the technical, economic, and societal challenges of advancing CCUS technologies and strategies at scale.

## Program Requirements

To receive the MSNT degree, students will need to complete at least 30 units of online coursework, which includes 9 units in core courses, including a 3-unit capstone course, and 21 credits of electives.

### Required Courses

| | | |
|---|---|---|
| CCUS520 | CLIMATE CHANGE AND SUSTAINABILITY | 3.0 |
| EBGN502 | POLITICAL ECONOMY OF THE ENERGY TRANSITION | 3.0 |
| CCUS585 | CARBON CAPTURE, UTILIZATION, AND STORAGE CAPSTONE | 3.0 |
| **Total Semester Hrs** | | **9.0** |

## Elective Courses

Choose 21 credits of coursework from those listed below. Other electives can be used with prior approval from the Program Director and your advisor.

| | | |
|---|---|---|
| CCUS521 | GEOLOGICAL CARBON CAPTURE UTILIZATION AND SEQUESTRATION (CCUS) | 3.0 |
| CCUS522 | NON-GEOLOGIC CARBON CAPTURE AND UTILIZATION | 3.0 |
| CCUS525 | BIOLOGICAL CARBON CAPTURE AND CONVERSION | 3.0 |
| CCUS530 | THE KINETICS OF CARBON DIOXIDE REACTIONS | 3.0 |
| CCUS598 | CLASS VI WELL DESIGN AND PERMITTING | 3.0 |
| GEOL557 | EARTH RESOURCE DATA SCIENCE 1: FUNDAMENTALS | 3.0 |
| GPGN519 | ADVANCED FORMATION EVALUATION | 3.0 |
| GPGN558 | SEISMIC DATA INTERPRETATION AND QUANTITATIVE ANALYSIS | 3.0 |
| PEGN506 | ENHANCED OIL RECOVERY METHODS | 3.0 |

## Carbon Capture, Utilization and Storage Certificate

The Graduate Certificate in Carbon Capture, Utilization and Storage (CCUS) is a fully online program designed to provide targeted, graduate-level training in the fundamentals of CCUS. It offers a flexible entry point for professionals or students seeking to build specialized expertise without committing to a full master's degree.

The certificate program requires three 3-credit graduate courses identified below: two required courses and the option to choose an elective course.

| | | |
|---|---|---|
| **Required Courses** | | |
| CCUS520 | CLIMATE CHANGE AND SUSTAINABILITY | 3.0 |
| EBGN502 | POLITICAL ECONOMY OF THE ENERGY TRANSITION | 3.0 |
| **Elective Course Options** | | |
| Choose one of the courses listed below. | | |
| CCUS521 | GEOLOGICAL CARBON CAPTURE UTILIZATION AND SEQUESTRATION (CCUS) | 3.0 |
| CCUS522 | NON-GEOLOGIC CARBON CAPTURE AND UTILIZATION | 3.0 |
| CCUS525 | BIOLOGICAL CARBON CAPTURE AND CONVERSION | 3.0 |
| CCUS530 | THE KINETICS OF CARBON DIOXIDE REACTIONS | 3.0 |
| CCUS598 | CLASS VI WELL DESIGN AND PERMITTING | 3.0 |
| PEGN506 | ENHANCED OIL RECOVERY METHODS | 3.0 |

**Course Modality:**

All courses are virtual and asynchronous - so students can listen to the lectures at convenience. All material is online and recorded for offline review. There are timed discussions and deliverables each week where students interact with their peers and with the instructors. The office hours are at specific times, but specific times to meet outside of those hours are often arranged depending on schedule conflicts or on geographic locations.

**CCUS520. CLIMATE CHANGE AND SUSTAINABILITY. 3.0 Semester Hrs.**

This eight-week online course is intended to introduce students to effects of atmospheric CO2 on climate, CO2 mitigation and avoidance strategies, and aspects of ESG when considering mitigation strategies. The course will provide students with much needed working knowledge about effects of Greenhouse Gases (GHGs) using data, and models. It provides cause and effects of GHGs as well as potential solutions that are equitable and sustainable.

**Course Learning Outcomes**

- 1) Use professional communication methods; explain in written format to a non-scientific reader (general public or policy maker) the big picture of Climate Change and what is Climate Change
- 2) Use professional communication methods; communicate in written format to a non-scientific reader (general public or policy maker) the big picture of Climate Change considering the role of Carbon in climate change
- 3) Using professional communication methods, explain to a non-scientific reader the big picture of Climate Change considering How can we assess climate change - what's the science (to the level needed to explain to general public) behind understanding the causes, and modeling for predictions
- 4) Explain in written format to a non-scientific reader (general public or policy maker) how scientists and engineers are studying solutions to climate change.
- 5) Analyze and accurately discuss the analysis of data sets related to climate change and CCUS considering data sources, data errors and uncertainties, and selecting accurate and appropriate data
- 6) Balance the atmospheric carbon budget in a proposed CCUS plan; consider CCUS with an ESG perspective
- 7) Discuss, using known scientific and social science perspectives, ethical considerations, societal impacts and issues of Climate Change and CCUS that should be considered as planning CCUS
- 8) Analyze maps of climate vulnerability; discuss climate equity; assess scientific accuracy of climate maps

**CCUS521. GEOLOGICAL CARBON CAPTURE UTILIZATION AND SEQUESTRATION (CCUS). 3.0 Semester Hrs.**

This course will cover sub-surface aspects of sustainable CCUS projects. Specifically, the topics covered will be geology of the subsurface appropriate for CCUS, how to create sustainable projects, the physics of CO2 transport, injection and storage it's their modeling studies, practical aspects of CO2 flooding, monitoring and verification methods including seismic, gravity and electromagnetic methods, and assessing CO2 capacity and migration. Each week of the course is taught by experts in the area from geology to engineering to geophysics and covers essential topics such as Class VI CCUS wells and EPA permitting, sustainable project development, to detailed physics such as CO2 phase and flow in the subsurface.

**Course Learning Outcomes**

**CCUS522. NON-GEOLOGIC CARBON CAPTURE AND UTILIZATION. 3.0 Semester Hrs.**

Global warming due to increase in concentration of greenhouse gases is arguably the biggest societal problem of our generation and it could pose existential threat to hundreds of millions of people all around the world. There is no unique strategy to address the threats of global warming. It is going to take a concerted effort leveraging all possible technologies

including renewable energies and carbon sequestration to develop solutions to this problem. This class is a part of the CCUS course which is geared towards exploring the feasibility of carbon removal for minimizing the global warming. This course specifically focuses on technologies for sequestering carbon dioxide from air or point sources and utilization and/or storage of the sequestered carbon Prerequisite: Basic understanding of engineering principles, basic knowledge of impact of greenhouse gases on global warming, ability to read research papers, ability to write reports.

**Course Learning Outcomes**

- Use professional communication methods; explain in written format to a non- scientific reader (general public or policy maker) the big picture of global warming due to increasing concentrations of carbon dioxide
- Use professional communication methods; communicate in written format to a non- scientific reader (general public or policy maker) the big picture of carbon sequestration for addressing global warming
- Using professional communication methods, explain to a non-scientific reader the big picture of carbon sequestration including understanding the broader context of how a multi-prong approach is necessary to address global warming and specifically what technologies are available for carbon sequestration and storage
- Explain in written format to a non-scientific reader (general public or policy maker) how scientists and engineers are developing solutions to climate change by sequestering carbon dioxide directly from air and/or from point sources
- Understand the challenges and limitations of sequestering carbon dioxide directly from air and from point sources including understanding the process broadly by looking at energy utilization and economics of the process
- Assess the potential role of sequestration in balancing the atmospheric carbon budget in a proposed CCUS plan
- Discuss, using known scientific and social science perspectives, ethical considerations, societal impacts and issues of Climate Change and carbon sequestration
- Analyze feasibility of various approaches discussed in this course and compare with systems already in place

**CCUS525. BIOLOGICAL CARBON CAPTURE AND CONVERSION. 3.0 Semester Hrs.**

Plants, bacteria, and algae have evolved over billions of years to efficiently use sunlight to turn carbon dioxide and water into useful chemicals, a potential solution to the very real problem of climate change. How do they do this? How can we take advantage of these processes? And, how can we improve on them to find practical solutions to address climate change? The purpose of this course is to answer these questions by introducing students to bioconversion, nature's ability to convert CO2 into other molecules. In this online, asynchronous course, students will work collaboratively in small teams to discuss current literature on carbon dioxide bioconversion, evaluate the effects of various parameters on bioconversion rates and efficiencies, and design and present case studies on ways to use plants, algae, and bacteria to convert CO2 to useful material. Prerequisite: None Co-requisite: None Prerequisite: Undergraduate introductory biology course (BIOL300 and BIOL301, or equivalent),Undergraduate general chemistry course (CHGN121 and CHGN122, or equivalent).

**Course Learning Outcomes**

- 1) Analyze how plants, algae, and bacteria contribute to the carbon cycle
- 2) Analyze how plants, algae, and bacteria contribute to the carbon cycle
- 3) Analyze how plants, algae, and bacteria contribute to the carbon cycle
- 4) Evaluate the effect various parameters (such as temperature, pH, salinity, light, etc.) have on the ability of bacteria to convert CO2 via photosynthesis
- 5) Recommend an appropriate biological-based method for capturing CO2 and converting it to a biofuel, based on the needs and available resources of a stakeholder
- 6) Demonstrate the ability to productively contribute to a team evaluating biological methods for CO2 capture and conversion
- 7) Develop and deliver a presentation on biological CO2 capture and conversion catalysts to an audience of postgraduate engineers

**CCUS530. THE KINETICS OF CARBON DIOXIDE REACTIONS. 3.0 Semester Hrs.**

Carbon dioxide is an extremely stable molecule that stays in the atmosphere for hundreds of years. What makes it so stable? Why does it take so long to convert to something else? How can we, as scientists and engineers, use chemistry to turn carbon dioxide into other, useful things, like fuels and materials? The purpose of this course is to answer these questions by delving into the thermodynamics of carbon dioxide, the kinetics of CO2 reactions, and how electrochemistry, photochemistry, and catalysts can overcome these problems. In this online, asynchronous course, students will work collaboratively in small teams to discuss current literature on CO2 conversion reactions, evaluate the pros and cons of electrochemistry and photochemistry for CO2 conversion, and design and present case studies on ways to use electrocatalysts and photocatalysts to convert CO2 to useful materials. Prerequisite: None Co-requisite: None Prerequisite: Undergraduate introductory chemistry (CHGN121 and CHGN122, or equivalent),Differential equations (MATH225 or equivalent).

**Course Learning Outcomes**

- 1. Propose electrochemical experiments, and interpret the resulting data, that would allow you to differentiate between CO2 reaction mechanisms
- 2. Propose photochemical experiments, and interpret the resulting data, that would allow you to differentiate between CO2 reaction mechanisms
- 3. Explain the underlying thermodynamic problems associated with converting CO2 into other carbon-based chemicals.
- 4. Evaluate CO2 conversion catalysts, based on various methods such as selectivity, turnover, activity, and cost
- 5. Evaluate CO2 reaction mechanisms based on experimental data
- 6. Develop and deliver a presentation on CO2 conversion catalysts to an audience of postgraduate engineers
- 7. Demonstrate the ability to productively contribute to a team evaluating CO2 conversion catalysts

**CCUS585. CARBON CAPTURE, UTILIZATION, AND STORAGE CAPSTONE. 3.0 Semester Hrs.**

In this course students will work individually and in groups to complete a project that requires them to integrate the skills and principles they have learned throughout the CCUS program curriculum and apply these skills to real systems. Prerequisites: CCUS520, EBGN502.

**Course Learning Outcomes**